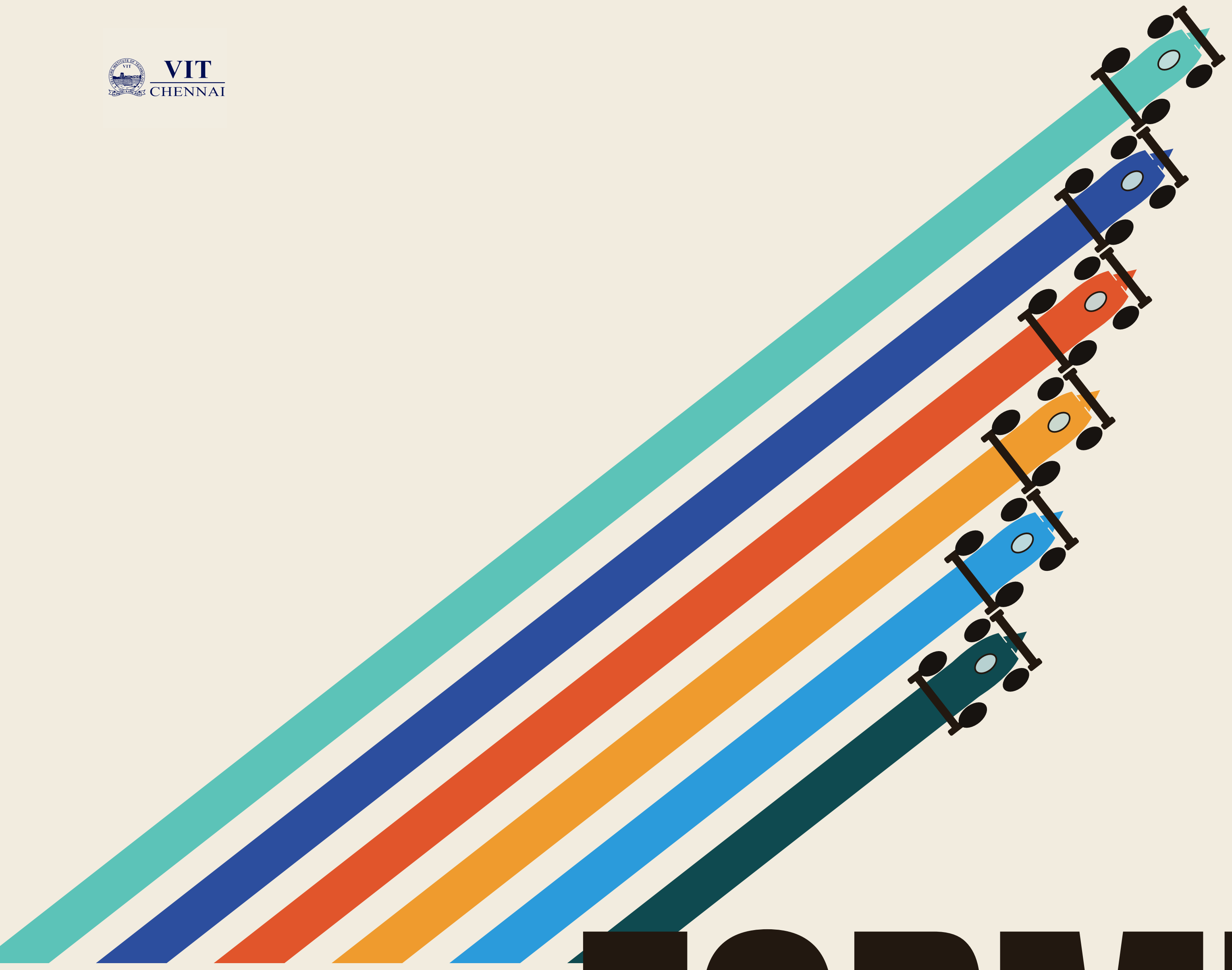
VIT CHENNAI

technoVIT



FORMULA1

WORKSHOP BROCHURE

SPEED · STRATEGY · PRECISION

A hands-on Reinforcement Learning workshop by the AI/ML Department — build and train an autonomous race car from scratch.

31.08.2026 ◆ AB3 — 401 ◆ 11 AM – 6 PM

CONTENTS

01 ABOUT THE WORKSHOP

Participants will learn the fundamentals of Reinforcement Learning through an interactive, hands-on workshop where they build and train an autonomous race car capable of navigating a racing track.

The workshop begins with the theoretical foundations of Reinforcement Learning, including agents, environments, states, actions, rewards, exploration versus exploitation, Markov Decision Processes, popular RL algorithms, and real-world applications.

In the second half, participants work independently to train their own autonomous racing AI inside a pre-built Gymnasium-based racing environment — covering observation space, action space, reward engineering, the training pipeline, model evaluation, and exporting trained weights.

The workshop concludes with a live AI Race Showcase where every participant's trained model drives on the same circuit.

NOT A COMPETITION

There are no winners, no rankings, no elimination, and no prizes. The showcase exists purely for learning — to demonstrate Reinforcement Learning in action and observe how differently trained models behave on the same track.

02 OBJECTIVES

This workshop is designed to:

- Introduce participants to the core theoretical foundations of Reinforcement Learning.
- Provide hands-on experience training a real RL agent from start to finish.
- Demonstrate how reward engineering and training parameters shape agent behavior.
- Give participants practical exposure to an end-to-end RL workflow.
- Showcase Reinforcement Learning in action through a shared race demonstration.

THINK → STRATEGIZE → SOLVE → COMPETE

03 WHO SHOULD ATTEND

Open to TechnoVIT participants with an interest in Artificial Intelligence and Machine Learning, particularly:

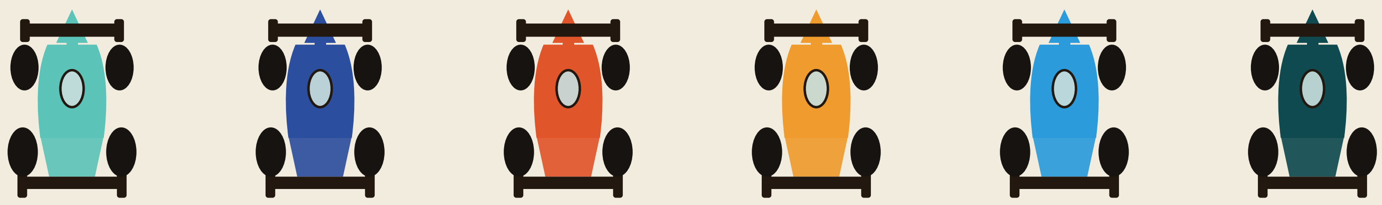
- Students curious about Reinforcement Learning and how AI agents learn through interaction.
- Participants who enjoy hands-on, project-based learning over purely theoretical sessions.
- Individuals looking to build and showcase a working AI project within a single day.

04 PREREQUISITES

No prior experience with Reinforcement Learning is required. Participants should have:

- Basic knowledge of Python programming.
- Familiarity with running Python scripts and installing packages (helpful, not mandatory).
- An interest in AI/ML concepts and a willingness to experiment and learn independently.

> **Test your logic, challenge your problem-solving skills, and see if you have what it takes to take on the grid.**
>
> — FORMULA1 WORKSHOP



SIX LANES. ONE TRACK. EVERY MODEL RACES THE SAME CIRCUIT.

05 HOW IT WORKS

FormulA1 is an individual event — every driver builds and races solo:

- Open to individual participants — no team registration required.
- Each participant works through the full pipeline solo: theory, training, and the race showcase.
- Training happens inside a pre-built Gymnasium racing environment, provided at the workshop.
- Mentors circulate throughout the room to help debug, tune, and speed up training.

BUILT ON GYMNASIUM

The hands-on session runs entirely inside Gymnasium, the standard open-source toolkit for developing and comparing reinforcement learning agents — if you've used OpenAI Gym before, it will feel immediately familiar.

06 WHAT TO BRING

Everything each driver needs before rolling out of the garage:

- [x] Laptop, capable of running Python and the training environment.
- [x] Python installed and working prior to the workshop.
- [x] Basic Python knowledge.

COME WITH PYTHON ALREADY INSTALLED — THE WORKSHOP BEGINS AT 11:00 AM.

07 RACE DAY SCHEDULE

A single, structured race day — from lights-out to the final lap.

11:00
11:15 AM

WELCOME & INTRODUCTION

About MIC AI/ML Department

Workshop objectives

Event overview

11:15 AM
1:00 PM

THEORY + PRACTICAL SESSION

Theory: agent, environment, state, action, reward · exploration vs. exploitation · Markov Decision Process · RL algorithms & applications

Practical: understanding the Gymnasium racing environment · observation & action space · reward function & training pipeline · demonstration

1:00
1:45 PM

LUNCH BREAK

Pit stop — refuel and recharge

1:45
4:45 PM

HANDS-ON TRAINING SESSION

Configure the Gymnasium environment, train the RL agent, monitor training progress

Test the autonomous vehicle and export model weights — mentors assist throughout

4:45
5:00 PM

MODEL SUBMISSION

Upload weights, validate submissions, load every participant's model

5:00
5:45 PM

AI RACE SHOWCASE — FINAL LAP

Every trained AI driver races the same circuit

Different strategies, behaviors & reward optimization on display — a showcase, not a competition

5:45
6:00 PM

CLOSING SESSION

Discussion & key takeaways, Q&A, feedback, group photograph

08 WHAT YOU'LL GAIN

By the end of the workshop, every participant's telemetry should read:

01 RL FUNDAMENTALS
A solid understanding of Reinforcement Learning fundamentals.

02 AGENT TRAINING
Hands-on experience building and training an RL agent.

03 REWARD ENGINEERING
Practical knowledge of how reward design shapes agent behavior.

04 TRAINING WORKFLOW
Understanding of complete RL training workflows.

05 AUTONOMOUS DECISIONS
First-hand observation of autonomous decision-making in action.

06 END-TO-END PROJECT
Experience with a full, end-to-end Reinforcement Learning project.

09 WHAT YOU LEAVE WITH

At the checkered flag, each participant drives away with:

- [x] A trained Reinforcement Learning agent capable of navigating the racing environment.
- [x] Exported model weights, submitted for the closing showcase.
- [x] Practical familiarity with the full training pipeline used to produce the agent.

EVERY PARTICIPANT LEAVES THE GARAGE WITH A WORKING, TRAINED AI RACER.

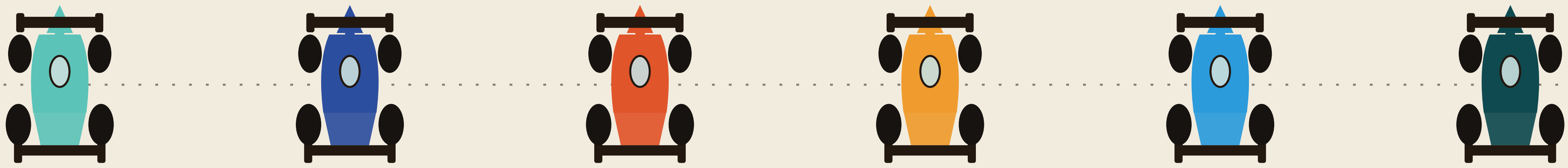
10 THE FINAL LAP

Every trained AI driver takes to the same circuit for one final lap. The showcase demonstrates:

- Different learning strategies and driving behaviors.
- Agent consistency under the same track conditions.
- Reward optimization in practice.
- Reinforcement Learning, in action, from grid to finish line.

SHOWCASE, NOT A COMPETITION

There are no rankings, no elimination, and no prizes. The Final Lap is a shared celebration of what every participant built — not a race to be won.



GRID TO FINISH LINE

SEE YOU ON THE GRID.

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